Conjugacy Classes, Normal Subgroups and Quotient Groups

Week 3 · Lecture 5

Mehmet Özkan

Group Theory

Reading for Lecture 5

- **Zee, I.1, pp. 44–45:** direct products, which we did not reach in Lecture 4.
- **Zee, I.2, p. 58, p. 62 and pp. 64–65:** conjugation, invariant subgroups, kernels and quotient groups.
- **Bünemann, §2.3.2, pp. 18–19 and §2.3.5–2.3.6, pp. 22–24:** classes, normal subgroups and factor groups.

We conjugate as Zee does, $g^{-1}ag$. Zee says *invariant subgroup* where we say normal subgroup; Bünemann also calls a quotient group a *factor group*. Products act from right to left.

Group Theory

Direct products of groups (from Lecture 4)

The direct product $G \times K$ consists of pairs (g, k) with $g \in G$ and $k \in K$.
Multiplication is componentwise:

$$(g_1, k_1)(g_2, k_2) = (g_1 g_2,\ k_1 k_2).$$

Closure and associativity hold in each component. The identity and inverses are

$$e_{G\times K} = (e_G, e_K), \qquad (g, k)^{-1} = (g^{-1}, k^{-1}).$$

For finite groups, $|G \times K| = |G|\,|K|$.

Textbook: Zee, I.1, pp. 44–45; Zee uses $\otimes$ for this group product.

Group Theory

$Z_2 \times Z_3$ is cyclic

Define $\psi : Z_6 \to Z_2 \times Z_3$ by

$$\psi([k]_6) = ([k]_2, [k]_3).$$

It is well defined because 6 is divisible by both 2 and 3, and it preserves addition componentwise. If $\psi([k]_6) = (0, 0)$, then k is divisible by both 2 and 3, hence by 6. Therefore $\ker \psi = \{0\}$, so ψ is injective. Both groups have six elements, so ψ is also surjective:

$$Z_6 \cong Z_2 \times Z_3.$$

The element $(1, 1) = \psi([1]_6)$ generates the product group.

Group Theory

$Z_2 \times Z_2$ and the group D_2

The product has four elements: $(0,0)$, $(1,0)$, $(0,1)$, $(1,1)$. Bünemann's D_2 consists of the identity and the half-turns about three perpendicular axes:

$$D_2 = \{e, u, v, uv\}, \qquad u^2 = v^2 = e, \quad uv = vu.$$

Define $\theta : Z_2 \times Z_2 \to D_2$ by $\theta(a,b) = u^a v^b$. The relations make this well defined and multiplication preserving. It maps the four pairs to four distinct elements, so

$$Z_2 \times Z_2 \cong D_2.$$

Textbook: Zee calls this abstract group V: I.1, p. 45. Compare Bünemann, §2.2.1.

Group Theory

Why $Z_2 \times Z_2$ is not Z_4

Every nonidentity element of $Z_2 \times Z_2$ has order 2:

$$(a, b) + (a, b) = (0, 0).$$

But Z_4 contains elements of order 4, such as $[1]_4$. An isomorphism would preserve element orders. Therefore

$$Z_2 \times Z_2 \ncong Z_4.$$

Having the same number of elements does not make two groups isomorphic. Their multiplication structure matters. We meet V again in Lecture 6, inside the rotation group of a tetrahedron.

Group Theory

Conjugation

Two elements $a, b \in G$ are *conjugate* if

$$b = g^{-1} a\, g \qquad \text{for some } g \in G.$$

In D_3, $srs = r^{-1} = r^2$, so r and r^2 are conjugate. Conjugation describes the same kind of transformation after a change of labels or of symmetry axis: a counterclockwise turn, seen in a mirror, is a clockwise turn.

Textbook: Zee, I.2, p. 58; Bünemann, §2.3.2.

Group Theory

Conjugacy classes

Write $a \sim b$ when a and b are conjugate. This is an equivalence relation:

- Reflexive: $a = e^{-1} a e$.
- Symmetric: $b = g^{-1} a g$ implies $a = g b g^{-1} = (g^{-1})^{-1} b\, g^{-1}$.
- Transitive: if $b = g^{-1} a g$ and $c = h^{-1} b h$, then $c = (gh)^{-1} a\, (gh)$.

The class containing a is

$$\mathrm{Cl}(a) = \{\, g^{-1} a g : g \in G \,\}.$$

Distinct classes are disjoint and together cover G. Unlike cosets, their sizes need not be equal.

Group Theory

First properties of classes

Since $g^{-1}eg = e$, the identity forms a class by itself. No other class is a subgroup, because no other class contains e. In an Abelian group $g^{-1}ag = a$, so every element is its own class: Z_3 has the classes $\{e\}$, $\{r\}$, $\{r^2\}$.

Conjugate elements have the same order, since

$$(g^{-1}ag)^m = g^{-1}a^m g,$$

which equals e exactly when $a^m = e$. The converse fails: in $Z_2 \times Z_2$ all three nonidentity elements have order 2, but each is its own class.

Group Theory

The three classes of D_3

Using $sr = r^{-1}s$ and $r^3 = e$:

$$s^{-1}rs = r^2, \qquad r^{-1}sr = rs, \qquad r^{-1}(rs)r = r^2 s, \qquad r^{-1}(r^2 s)r = s.$$

Conjugation by s exchanges r and r^2; conjugation by r cycles the three reflections. Conjugates have the same order, so rotations and reflections never mix:

$$\{e\}, \qquad \{r,\, r^2\}, \qquad \{s,\, rs,\, r^2 s\}, \qquad 1 + 2 + 3 = 6.$$

In S_3 these are $\{e\}$, the two 3-cycles and the three transpositions.

Textbook: In Bünemann's prism realization the last class consists of three twofold rotations.

Group Theory

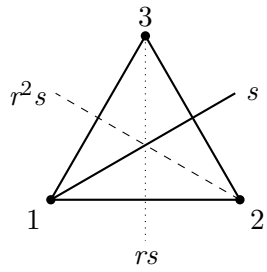
Classes as geometry

As permutations of the vertex labels, $r = (123)$ and $s = (23)$. Conjugating s by a rotation moves its mirror to another mirror:

$$r^{-1}sr = rs = (12),$$

the reflection in the mirror through vertex 3. Any mirror can be turned into any other, so the three reflections form one class. The sign χ of Lecture 3 takes one value on each class:

$$\chi(g^{-1}ag) = \chi(g)^{-1}\chi(a)\chi(g) = \chi(a).$$



Group Theory

Exercise: the classes of D_4

Label the vertices of the square $1, 2, 3, 4$ counterclockwise. Then $r = (1234)$ and $s = (24)$, the reflection in the diagonal through vertex 1:

$$D_4 = \langle r, s \mid r^4 = e,\ s^2 = e,\ srs = r^{-1} \rangle.$$

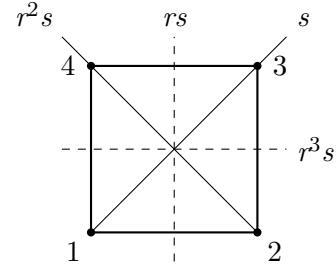
1. Find all conjugacy classes of D_4. Use $s\,r^k = r^{-k}s$.
2. Which reflections are conjugate to s? Draw their mirror lines.
3. Check that the class sizes add up to 8.

Group Theory

Solution: five classes

$$\{e\}, \quad \{r^2\}, \quad \{r, r^3\}, \quad \{s, r^2 s\}, \quad \{rs, r^3 s\}.$$

The half-turn r^2 commutes with every element. The diagonals are $s = (24)$ and $r^2 s = (13)$; the mirrors of $rs = (12)(34)$ and $r^3 s = (14)(23)$ join midpoints of opposite edges. No symmetry of the square carries a diagonal to an edge line. In D_3 every mirror passes through a vertex, so all three are conjugate.



Group Theory

Normal subgroups

For $g \in G$, the conjugate set $g^{-1}Hg = \{g^{-1}hg : h \in H\}$ is again a subgroup. A subgroup $H \leq G$ is *normal*, written $H \trianglelefteq G$, if

$$g^{-1}Hg = H \qquad \text{for every } g \in G.$$

Zee calls this an *invariant subgroup*. The equality concerns the whole set; individual elements of H may move. Multiplying on the left by g gives an equivalent condition:

$$H \trianglelefteq G \iff gH = Hg \quad \text{for every } g \in G.$$

Textbook: Zee, I.2, p. 62; Bünemann, §2.3.5.

Group Theory

Normal subgroups and whole classes

If $H \trianglelefteq G$ and $h \in H$, then $g^{-1}hg \in H$ for every g: the whole class of h lies in H. Conversely, if a subgroup contains the class of each of its elements, then $g^{-1}Hg \subseteq H$ for every g; using g^{-1} in place of g gives $H \subseteq g^{-1}Hg$. Hence

$$H \trianglelefteq G \iff H \text{ is a union of complete conjugacy classes.}$$

To disprove normality, one pair $g \in G$, $h \in H$ with $g^{-1}hg \notin H$ is enough.

Textbook: Bünemann, §2.3.5, p. 23.

Group Theory

A union of classes still needs closure

In D_3, the set

$$\{e,\ s,\ rs,\ r^2 s\}$$

is a union of complete classes. But it is not a subgroup:

$$s(rs) = srs = r^2$$

lies outside the set. The normality criterion applies to a set already known to be a subgroup.

Group Theory

Exercise: normal subgroups of D_3

The classes have sizes 1, 2 and 3:

$$\{e\}, \qquad \{r, r^2\}, \qquad \{s, rs, r^2 s\}.$$

1. Which unions of classes contain e?
2. Which can be subgroups according to Lagrange's theorem?
3. List every normal subgroup of D_3.

Group Theory

Exercise: solution

The possible union sizes are 1, 3, 4 and 6. Size four cannot be a subgroup because $4 \nmid 6$. The remaining sets are subgroups:

$$\{e\}, \qquad \{e, r, r^2\}, \qquad D_3.$$

These are exactly the normal subgroups. The same result applies to $S_3 \cong D_3$.

In particular $H = \{e, s\}$ is not normal: $r^{-1}sr = rs \notin H$. Equivalently, as we found in Lecture 4, $rH = \{r, rs\} \neq Hr = \{r, r^2 s\}$.

Group Theory

Three sources of normal subgroups

- $\{e\}$ and G itself are always normal.
- In an Abelian group every subgroup is normal, since $g^{-1}hg = h$.
- A subgroup of index 2 is normal.

Proof of the last statement: if $[G : H] = 2$, the two left cosets are H and its complement $G \setminus H$. So are the two right cosets. For $g \in H$, $gH = Hg = H$; for $g \notin H$,

$$gH = Hg = G \setminus H.$$

This gives $\langle r \rangle \trianglelefteq D_3$ without computing a single conjugate.

Group Theory

Every kernel is normal

Let $\varphi : G \to K$ be a homomorphism and $N = \ker \varphi$. For $n \in N$ and any $g \in G$,

$$\varphi(g^{-1}ng) = \varphi(g)^{-1}\varphi(n)\varphi(g) = \varphi(g)^{-1}e_K\varphi(g) = e_K.$$

Thus $g^{-1}Ng \subseteq N$. Applying the same argument with g^{-1} gives the reverse inclusion:

$$\ker \varphi \trianglelefteq G.$$

Examples from Lecture 3: $\ker \chi = \langle r \rangle \trianglelefteq D_3$ and $\ker(Z_6 \to Z_3) = \{0, 3\} \trianglelefteq Z_6$.

Textbook: Zee, I.2, p. 64.

Group Theory

Multiplying cosets

For $N \trianglelefteq G$ we want $(aN)(bN) = (ab)N$. A coset has many representatives, so the answer must depend only on the cosets. Write other representatives as $a' = an_1$ and $b' = bn_2$ with $n_1, n_2 \in N$:

$$a'b' = a\, n_1\, b\, n_2 = ab\,(b^{-1} n_1 b)\, n_2.$$

Normality gives $b^{-1} n_1 b \in N$, so $a'b' = (ab)n_3$ with $n_3 \in N$, and $(a'b')N = (ab)N$. We moved n_1 past b by conjugation, without assuming that they commute.

Textbook: Zee, I.2, Eq. (9), pp. 64–65; Bünemann, §2.3.6.

Group Theory

Why normality is needed

Try the same rule with $H = \{e, s\} \leq D_3$, which is not normal. The elements r and rs label the same coset $rH = \{r, rs\}$. Multiply each by r:

$$r \cdot r = r^2 \in r^2 H, \qquad (rs) \cdot r = r\,(sr) = r\,r^{-1} s = s \in H.$$

The “product” $rH \cdot rH$ would be both $r^2 H$ and H. The proposed product of cosets is not well defined.

Group Theory

The quotient group

For $N \trianglelefteq G$, the distinct cosets form the group

$$G/N = \{gN : g \in G\}, \qquad (aN)(bN) = (ab)N.$$

Zee calls it a *quotient group*; Bünemann also calls it a *factor group*.

- Associativity: both ways of multiplying three cosets give $abcN$.
- Identity: $N = eN$. Inverse: $(aN)^{-1} = a^{-1}N$.

For finite groups, $|G/N| = [G : N] = |G|/|N|$. If $a \notin N$, then $e \notin aN$: such a coset is not a subgroup of G, but it is an element of G/N.

Group Theory

$D_3/\langle r\rangle$

With $N = \{e, r, r^2\}$, there are only two cosets:

$$N, \qquad sN = \{s, rs, r^2 s\}.$$

Two reflections compose to a rotation, so $sN \cdot sN = N$:

	N	sN
N	N	sN
sN	sN	N

$$D_3/\langle r\rangle \cong Z_2.$$

The sign χ of Lecture 3 is constant on each coset, so it induces a map on the quotient, $\bar{\chi}(gN) = \chi(g)$, with $\bar{\chi}(N) = +1$ and $\bar{\chi}(sN) = -1$. The quotient remembers only whether a symmetry reverses orientation.

Group Theory

A quotient in additive notation

In Z_6, let $N = \{0, 3\}$. Since Z_6 is Abelian, N is normal. The quotient has three elements,

$$N = \{0, 3\}, \qquad 1 + N = \{1, 4\}, \qquad 2 + N = \{2, 5\},$$

added by $(a + N) + (b + N) = (a + b) + N$. The coset $1 + N$ generates it:

$$1 + N, \quad 2 + N, \quad 3 + N = N.$$

So $Z_6/\{0, 3\} \cong Z_3$. The reduction map $Z_6 \to Z_3$ of Lecture 3 has kernel $\{0, 3\}$: the quotient groups together exactly the elements with the same image.

Group Theory

The canonical quotient map

The map $\pi : G \to G/N$, defined by $\pi(g) = gN$, is surjective and preserves multiplication:

$$\pi(ab) = abN = (aN)(bN) = \pi(a)\pi(b).$$

Moreover,

$$\pi(g) = N \iff g \in N.$$

Thus $\ker \pi = N$. Every element of N maps to the quotient identity, and every normal subgroup is the kernel of some homomorphism.

Group Theory

Kernels and quotient groups

For a homomorphism $\varphi : G \to K$ with $N = \ker \varphi$, define

$$\bar{\varphi} : G/N \to \operatorname{im} \varphi, \qquad \bar{\varphi}(gN) = \varphi(g).$$

The key chain is $\varphi(a) = \varphi(b) \iff \varphi(a^{-1}b) = e_K \iff a^{-1}b \in N \iff aN = bN$.

- *Well defined* ($\Leftarrow$): if $aN = bN$, then $\varphi(a) = \varphi(b)$.
- *Injective* ($\Rightarrow$): if $\varphi(a) = \varphi(b)$, then $aN = bN$.
- *Homomorphism*: $\bar{\varphi}(aN\,bN) = \varphi(ab) = \varphi(a)\varphi(b)$. *Onto* $\operatorname{im} \varphi$ by construction.

Hence $G/\ker \varphi \cong \operatorname{im} \varphi$: the *first isomorphism theorem*.

Group Theory

A physical quotient: periodic boundary conditions

Translations of an infinite chain by whole numbers of lattice spacings form the group $\mathbb{Z}$ under addition. A ring of N sites is the same chain with a translation by N sites declared to do nothing. Since $\mathbb{Z}$ is Abelian, $N\mathbb{Z} = \{Nk : k \in \mathbb{Z}\}$ is normal, and

$$\mathbb{Z}/N\mathbb{Z} \cong Z_N.$$

This is the shift group of the ring from Week 1. Periodic boundary conditions in a crystal replace the infinite translation group by a finite quotient of this kind.

Group Theory

Check your understanding: two quotients of order four

1. Let $N = \{0, 4\} \leq Z_8$. Explain why N is normal, list its cosets, identify Z_8/N, and find the order of $2 + N$.
2. Let $M = \{e, r^2\} \leq D_4$. Explain why M is normal, list its cosets, and decide which group of order four D_4/M is.

Hint: compute the square of each coset, and recall why $Z_2 \times Z_2 \ncong Z_4$.

Group Theory

Check your understanding: solution

1. Z_8 is Abelian. The cosets are $\{0,4\},\{1,5\},\{2,6\},\{3,7\}$, and they add like residues modulo four: $Z_8/N \cong Z_4$. The coset $2+N$ has order two, since $(2+N)+(2+N)=4+N=N$.
2. $M=\{e\}\cup\{r^2\}$ is a union of classes. The cosets are

$$M, \quad rM=\{r,r^3\}, \quad sM=\{s,r^2s\}, \quad rsM=\{rs,r^3s\}.$$

   Every coset squares to M: $(rM)^2=r^2M=M$, and s, rs have order 2. So $D_4/M \cong Z_2 \times Z_2$.

Two quotients of the same order can be different groups.

Group Theory

Lecture 5 in brief

- Conjugate elements $g^{-1}ag$ are the same kind of operation. Classes partition G and preserve element order.
- D_3: $\{e\}, \{r, r^2\}, \{s, rs, r^2s\}$. D_4: $\{e\}, \{r^2\}, \{r, r^3\}, \{s, r^2s\}, \{rs, r^3s\}$.
- Normal subgroups are unions of classes; kernels and subgroups of index two are normal.
- Normality makes $(aN)(bN) = abN$ well defined. The quotient identity is N, and $G/\ker\varphi \cong \operatorname{im}\varphi$.

Next: permutations, parity, the alternating group and the classes of S_n.

Reading: Zee, I.1, pp. 44–45 and I.2, pp. 58, 62–65; Bünemann, §2.3.2 and §2.3.5–2.3.6.

Permutations, Parity and the Alternating Group

Week 3 · Lecture 6

Mehmet Özkan

Group Theory

Reading for Lecture 6

- **Zee, I.2, pp. 55–60:** permutations, cycles, transpositions, parity, classes and cycle types.
- **Zee, I.2, pp. 59 and 62:** the Klein four-group inside A_4.
- **Bünemann, §2.2.1:** the group D_2, which reappears inside A_4.

We continue with e, Z_n and right-to-left composition. Cycles of length one are omitted. The A_4 slides at the end are optional enrichment.

Group Theory

The permutation group S_n

S_n consists of the bijections of $\{1, \ldots, n\}$ onto itself, with composition as multiplication. There are n choices for the image of 1, then $n-1$ choices for the image of 2, and so on:

$$|S_n| = n! \, .$$

A permutation is a map. Its notation must specify where each label goes.

Group Theory

Two-line notation and cycle notation

$$p = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (123).$$

The cycle means

$$1 \longmapsto 2 \longmapsto 3 \longmapsto 1.$$

Starting at another label does not change the arrows:

$$(123) = (231) = (312).$$

The bracketed list $[2, 3, 1]$ records $[p(1), p(2), p(3)]$ in the fixed input order.

Group Theory

Applying the cycle twice

For $p = (123)$,

$$p^2 : \quad 1 \mapsto 3, \quad 2 \mapsto 1, \quad 3 \mapsto 2.$$

Hence

$$p^2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = (132).$$

The image list is $[3, 1, 2]$. But the cycle (312) still equals (123). An image list and a cycle are different notations.

Group Theory

Images and occupants of positions

Initially, positions $1, 2, 3$ contain objects A, B, C. Let $p = (123)$ move each object to the image of its position.

Position	1	2	3
Before	A	B	C
After p	C	A	B
After p^2	B	C	A

The image list is $[2, 3, 1]$, but the new occupants are $[C, A, B]$. The object arriving at position j came from position $p^{-1}(j)$.

Group Theory

Right-to-left composition

For $(12)(23)$, apply (23) first:

x	1	2	3
$(23)(x)$	1	3	2
$(12)\big((23)(x)\big)$	2	3	1

$$\implies \qquad (12)(23) = (123).$$

In the other order the arrows are $1 \mapsto 3$, $2 \mapsto 1$, $3 \mapsto 2$, so

$$(23)(12) = (132) \neq (123) = (12)(23).$$

Overlapping cycles can fail to commute. Disjoint cycles commute because they act on separate labels.

Group Theory

Disjoint-cycle decomposition

Consider Zee's example:

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 1 & 5 & 2 & 3 \end{pmatrix}.$$

Follow the arrows: $1 \to 4 \to 2 \to 1$ and $3 \to 5 \to 3$. Thus $g = (142)(35)$.

Every permutation decomposes this way: repeat the procedure on unused labels. A label must return to itself, because $p^i(a) = p^j(a)$ with $i < j$ gives $p^{j-i}(a) = a$. The decomposition is unique up to the order of the cycles and the starting point inside each cycle.

Textbook: Zee, I.2, p. 56.

Group Theory

Inverses and orders of permutations

The inverse reverses each cycle:

$$(142)^{-1} = (124), \qquad \big((142)(35)\big)^{-1} = (124)(35).$$

For disjoint cycles of lengths $k_1, \ldots, k_t$,

$$\operatorname{ord}(p) = \operatorname{lcm}(k_1, \ldots, k_t).$$

All cycles must return to their starting positions simultaneously. Thus $(142)(35)$ has order 6, while $(12)(34)$ has order 2. S_3 has no element of order 6; S_5 does.

Group Theory

Transpositions

A transposition exchanges two labels, such as (12). A k-cycle is a product of $k-1$ transpositions:

$$(a_1\, a_2 \cdots a_k) = (a_1\, a_2)(a_2\, a_3) \cdots (a_{k-1}\, a_k).$$

For example, $(142) = (14)(42)$ and $(142)(35) = (14)(42)(35)$. Thus every permutation is a product of transpositions. The product is not unique:

$$(123) = (12)(23) = (13)(12).$$

Textbook: Zee, I.2, pp. 56–57.

Group Theory

Even or odd?

Both ways of writing (123) above use two transpositions. Inserting $(12)(12) = e$ adds two more. Can the same permutation also be written with an *odd* number of transpositions?

To answer this we need a quantity that

- is unchanged by how we write p,
- multiplies when permutations are composed,
- equals -1 on every transposition.

Then $p = t_1 t_2 \cdots t_m$ forces the value $(-1)^m$, and the parity of m is fixed.

Group Theory

The sign of a permutation

Define

$$\operatorname{sgn}(p) = \prod_{i<j} \frac{p(j) - p(i)}{j - i}.$$

Each pair of labels appears exactly once in the numerator, possibly in the wrong order, so $\operatorname{sgn}(p) = \pm 1$. It is -1 to the number of pairs $i < j$ with $p(i) > p(j)$. In S_3:

$$\operatorname{sgn}\big((12)\big) = \frac{1-2}{1} \cdot \frac{3-2}{2} \cdot \frac{3-1}{1} = -1, \qquad \operatorname{sgn}\big((123)\big) = \frac{3-2}{1} \cdot \frac{1-2}{2} \cdot \frac{1-3}{1} = +1.$$

Group Theory

The sign is a homomorphism

For $p, q \in S_n$, insert $q(j) - q(i)$ above and below:

$$\operatorname{sgn}(pq) = \prod_{i<j} \frac{p(q(j)) - p(q(i))}{q(j) - q(i)} \prod_{i<j} \frac{q(j) - q(i)}{j - i}.$$

The second product is $\operatorname{sgn}(q)$. In the first, each factor $\frac{p(b)-p(a)}{b-a}$ is symmetric in a, b, and $\{q(i), q(j)\}$ runs over every pair once. So the first product is $\operatorname{sgn}(p)$:

$$\operatorname{sgn}(pq) = \operatorname{sgn}(p) \operatorname{sgn}(q).$$

Thus $\operatorname{sgn} : S_n \to \{+1, -1\}$ is a homomorphism.

Group Theory

How the pairs are rearranged: an example

Take $q = (123) \in S_3$. The three pairs $i < j$ are sent to

(i, j)	$(1,2)$	$(1,3)$	$(2,3)$
$(q(i), q(j))$	$(2,3)$	$(2,1)$	$(3,1)$

As unordered pairs these are $\{2,3\}$, $\{1,2\}$, $\{1,3\}$: each pair appears exactly once. Only the order inside a pair can flip, as in $(2,1)$. The factor

$$\frac{p(b) - p(a)}{b - a}$$

does not change when a and b are exchanged, so the flips do not matter. The first product in $\operatorname{sgn}(pq)$ therefore has the same factors as $\operatorname{sgn}(p)$.

Group Theory

Transpositions are odd

For $(12) \in S_n$, only the pair $(1, 2)$ changes order, so $\operatorname{sgn}\big((12)\big) = -1$. Any transposition is conjugate to it: $(ab) = f(12)f^{-1}$ with $f(1) = a$, $f(2) = b$. A homomorphism into an Abelian group takes one value on a class, so

$$\operatorname{sgn}\big((ab)\big) = \operatorname{sgn}(f)\,(-1)\,\operatorname{sgn}(f)^{-1} = -1.$$

Therefore, if $p = t_1 \cdots t_m$ with transpositions t_i, then $\operatorname{sgn}(p) = (-1)^m$. Different decompositions always have the same parity: p is *even* or *odd*.

Group Theory

The sign of a cycle

A k-cycle is a product of $k-1$ transpositions, so its sign is $(-1)^{k-1}$:

$$(123) \text{ is even}, \qquad (1234) \text{ is odd}, \qquad (12)(34) \text{ is even}.$$

The number of moved labels alone does not determine the sign. If $p \in S_n$ has c cycles, counting fixed labels as 1-cycles,

$$\operatorname{sgn}(p) = (-1)^{n-c}.$$

For Zee's example, $(142)(35)$ has sign $(+1)(-1) = -1$.

Group Theory

The alternating group A_n

The even permutations form the alternating group:

$$A_n = \{p \in S_n : \operatorname{sgn}(p) = +1\} = \ker(\operatorname{sgn}) \trianglelefteq S_n,$$

normal because every kernel is normal. For $n \geq 2$, fix a transposition t. The map $p \mapsto tp$ pairs every even permutation with an odd one, and multiplying by t again reverses the pairing. Thus exactly half the permutations are even:

$$|A_n| = \frac{n!}{2}, \qquad S_n/A_n \cong Z_2.$$

The two cosets are the even and the odd permutations.

Group Theory

S_3/A_3

$A_3 = \{e, (123), (132)\} \cong Z_3$. Its two cosets are

$$A_3 \qquad \text{and} \qquad (12)A_3 = \{(12), (23), (13)\}.$$

Under $D_3 \cong S_3$ this is $D_3/\langle r \rangle$ from Lecture 5: rotations of the triangle are even permutations of its vertices, and reflections are odd. The sign χ of Lecture 3 is the sign of a permutation:

$$\chi = \mathrm{sgn} \quad \text{on } D_3 \cong S_3.$$

Group Theory

Where the sign appears in physics

- The Levi-Civita symbol is the sign: $\varepsilon_{p(1)p(2)p(3)} = \operatorname{sgn}(p)$. The cyclic orders $123, 231, 312$ are the even permutations.
- The determinant is a signed sum over S_n:

$$\det M = \sum_{p \in S_n} \operatorname{sgn}(p)\, M_{1p(1)} M_{2p(2)} \cdots M_{np(n)}.$$

- The product $\Delta(x_1, \ldots, x_n) = \prod_{i<j}(x_j - x_i)$ is antisymmetric:

$$\Delta(x_{p(1)}, \ldots, x_{p(n)}) = \operatorname{sgn}(p)\, \Delta(x_1, \ldots, x_n).$$

★ For those who know quantum mechanics: wave functions of identical fermions change by $\operatorname{sgn}(p)$ under exchange.

Group Theory

Conjugation relabels cycles

Using fpf^{-1} instead of $f^{-1}pf$ gives the same set of conjugates. If $p = (a_1 \cdots a_k)$, then

$$fpf^{-1} = \big(f(a_1) \cdots f(a_k)\big).$$

Indeed, the arrows are

$$f(a_i) \xrightarrow{f^{-1}} a_i \xrightarrow{p} a_{i+1} \xrightarrow{f} f(a_{i+1}),$$

and a label not of the form $f(a_i)$ is fixed. For a product of cycles, $f(c_1 c_2)f^{-1} = (fc_1f^{-1})(fc_2f^{-1})$. Conjugation changes the labels while preserving the cycle lengths.

Group Theory

Conjugacy classes of S_3

For $f = (123)$,

$$f(12)f^{-1} = (23).$$

For $f = (23)$,

$$f(123)f^{-1} = (132).$$

Thus the three classes are

$$\{e\}, \qquad \{(123), (132)\}, \qquad \{(12), (13), (23)\}.$$

They match the three classes of D_3 from Lecture 5.

Group Theory

Cycle type and conjugacy in S_n

The *cycle type* lists the lengths of the disjoint cycles, including fixed points. For example, (123) in S_4 has type $(3, 1)$.

$$\text{Conjugacy in } S_n \iff \text{the same cycle type.}$$

Conjugation preserves the lengths. Conversely, matching labels along equal-length cycles constructs the relabelling:

$$(1\,2)(3\,4) \longrightarrow (1\,3)(2\,4): \qquad f: 1 \mapsto 1,\ 2 \mapsto 3,\ 3 \mapsto 2,\ 4 \mapsto 4, \quad f = (23).$$

The equivalence refers specifically to conjugacy in S_n.

Group Theory

Counting one cycle type

Let m_j be the number of j-cycles, so $\sum_j j\, m_j = n$. Then

$$\text{Class size in } S_n = \frac{n!}{\prod_j j^{m_j}\, m_j!}.$$

Start with $n!$ arrangements of labels. Divide by j for every cyclic choice of starting point, and by $m_j!$ for the ordering of equal-length cycles. For $(12)(34)$ in S_4, the size is $4!/(2^2\, 2!) = 3$.

Textbook: Zee, I.2, Eq. (4), pp. 59–60.

Group Theory

The five classes of S_4

Type	Representative	Size	Order	Sign
$(1,1,1,1)$	e	1	1	$+1$
$(2,1,1)$	(12)	6	2	-1
$(2,2)$	$(12)(34)$	3	2	$+1$
$(3,1)$	(123)	8	3	$+1$
(4)	(1234)	6	4	-1

The sizes sum to 24. The even types contain $1 + 3 + 8 = 12$ elements, giving $|A_4| = 12$. The number of classes of S_n is the number of partitions of n: 3 for S_3, 5 for S_4, 7 for S_5.

Group Theory

Check your understanding

Let $p = (13)(25)(46) \in S_6$.

1. Find p^{-1} and $\mathrm{ord}(p)$.
2. Determine $\mathrm{sgn}(p)$.
3. Find the size of its conjugacy class in S_6.
4. Is p conjugate to $(123)(45)$ in S_6?

Hint: the cycle type is $(2, 2, 2)$.

Group Theory

Check your understanding: solution

The transpositions are disjoint, so

$$p^{-1} = p, \qquad p^2 = e, \qquad \operatorname{ord}(p) = 2.$$

There are three transpositions, giving $\operatorname{sgn}(p) = -1$. With $m_2 = 3$, the class size is

$$\frac{6!}{2^3\, 3!} = 15.$$

$(123)(45)$ has type $(3, 2, 1)$, so it is not conjugate to p. Both are odd, so the sign alone cannot tell them apart; their orders, 2 and 6, already do.

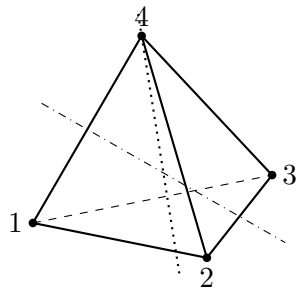
Group Theory

Optional: A_4 as the rotations of a regular tetrahedron

Label the vertices $1, 2, 3, 4$. Each rotation permutes them:

- $\pm 120^\circ$ about an axis through a vertex and the opposite face: $4 \times 2 = 8$ three-cycles, e.g. (123);
- 180° about an axis joining midpoints of opposite edges: 3 elements, e.g. $(14)(23)$;
- the identity.

$1 + 8 + 3 = 12 = 4!/2$, all even: the rotation group is A_4. Adding reflections gives all of S_4.



dotted: axis of (123)
dash-dot: axis of $(14)(23)$

Group Theory

Optional: the Klein four-group in A_4

Let

$$V = \{e,\ (12)(34),\ (13)(24),\ (14)(23)\}.$$

Each nonidentity element squares to e, and the product of two distinct ones is the third, for example $(12)(34)(13)(24) = (14)(23)$. Thus $V \leq A_4$ and $V \cong Z_2 \times Z_2$. For a regular tetrahedron its three half-turn axes are mutually perpendicular: V is Bünemann's D_2, sitting inside the tetrahedron.

Conjugation preserves the double-transposition type, so $V = \{e\} \cup \{\text{type } (2,2)\}$ is a union of classes, and $V \trianglelefteq A_4$.

Textbook: Zee, I.2, pp. 59 and 62; Bünemann, §2.2.1.

Group Theory

Optional: the quotient A_4/V

V has index $12/4 = 3$, and every group of prime order is cyclic, so

$$A_4/V \cong Z_3.$$

The three cosets are V and

$$(123)V = \{(123), (134), (243), (142)\}, \qquad (132)V = \{(132), (143), (234), (124)\}.$$

Every rotation of the regular tetrahedron permutes the three perpendicular half-turn axes. The elements of V keep each axis in place; the 3-cycles permute the axes cyclically. The quotient remembers only this.

Group Theory

Optional: A_4 has no subgroup of order six

Lagrange allows subgroups of order $1, 2, 3, 4, 6, 12$. Suppose $H \leq A_4$ with $|H| = 6$.

- Then $[A_4 : H] = 2$, so $H \trianglelefteq A_4$ and $A_4/H \cong Z_2$.
- Every element of Z_2 squares to the identity: $(gH)^2 = g^2H = H$, so $g^2 \in H$ for every $g \in A_4$.
- A 3-cycle c has $c^3 = e$, so $c = c^4 = (c^2)^2 \in H$.

All eight 3-cycles would lie in H, which has only six elements. The converse of Lagrange's theorem is false.

Group Theory

Optional: counting a class with the centralizer

The *centralizer* $C_G(a) = \{g \in G : ga = ag\}$ is a subgroup of G. Since the class is the same set whichever way we conjugate, count the conjugates gag^{-1}:

$$gag^{-1} = hah^{-1} \iff h^{-1}g\,a = a\,h^{-1}g \iff h^{-1}g \in C_G(a) \iff gC_G(a) = hC_G(a).$$

Distinct conjugates correspond to distinct cosets of $C_G(a)$, so

$$|\mathrm{Cl}_G(a)| = [G : C_G(a)] = \frac{|G|}{|C_G(a)|}.$$

In particular, class sizes divide $|G|$. In D_3: $C_{D_3}(r) = \{e, r, r^2\}$ gives $6/3 = 2$, and $C_{D_3}(s) = \{e, s\}$ gives $6/2 = 3$.

Group Theory

Optional: classes can split in a subgroup

All eight 3-cycles form one class in S_4, so $|C_{S_4}((123))| = 24/8 = 3$ and $C_{S_4}((123)) = \{e, (123), (132)\}$. In A_4 we may conjugate only by even permutations, so the relevant centralizer is $C_{A_4}((123)) = C_{S_4}((123)) \cap A_4$. Here the two coincide, since all three elements are even. Hence $|\mathrm{Cl}_{A_4}((123))| = 12/3 = 4$, and the 3-cycles split:

$$\{(123), (142), (134), (243)\}, \qquad \{(132), (124), (143), (234)\}.$$

For example, $(23)(123)(23) = (132)$, but $(23) \notin A_4$. Conjugacy must use elements of the group under discussion. A_4 has four classes, of sizes $1, 3, 4, 4$.

Textbook: Zee, I.2, p. 59.

Group Theory

Week 3 in brief

- Classes partition a group and preserve element order. Normal subgroups are unions of classes; G/N is a group, and $G/\ker\varphi \cong \operatorname{im}\varphi$.
- Disjoint cycles give permutation orders through a least common multiple.
- $\operatorname{sgn} : S_n \to \{\pm 1\}$ is a homomorphism, so parity is well defined. $A_n = \ker(\operatorname{sgn}) \trianglelefteq S_n$ and $S_n/A_n \cong Z_2$.
- Cycle type determines conjugacy in S_n.

Next week: a linear algebra bridge. Conjugation returns as change of basis, $M \mapsto S^{-1}MS$, and the trace will turn out to be constant on classes.

Reading: Zee, I.2, pp. 55–60; Bünemann, §2.2.1. Optional: Zee, I.2, pp. 59 and 62.